

A Take on Credence, Belief and Sleeping Beauty

based on John MacFarlane's recent paper "Belief: What is it Good for?"

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MacFarlane's Account

- I take him to be taking a methodologically behaviourist perspective on the concepts of "belief" and "credence": They are theoretical terms which have to be introduced in terms of observable behaviour.
- Then he asks: Which talents do they, respectively, bring to the table, and are both concepts' talents needed when we theorise human beings as rational and social beings?
 - **Credence**: Good at decision-theoretically rationalising and explaining an individual's actions. Belief is much worse at this.
 - **Belief** brings to the table
 - * "truth" as a correctness condition and
 - * being required for the ascription of certain reactive attitudes (cf. Lara Buchak: you should not resent someone for stealing your phone if you don't believe they stole your phone)
- What do these talents qualify "belief" for?
 - MacFarlane: For describing and explaining the social practice of *reasoning*. Reasoning proceeds by argument, which works by inferring the truth of a conclusion from the truth of a small set of premises. Furthermore, what e.g. resenting someone requires is exactly a *reason* for resenting them.
- So: we ascribe **belief** that p to someone iff p is a candidate reason for them; if they would use p as a reason in the social practice of reasoning.
- And we ascribe **credence** c that p to someone iff it (decision-theoretically) explains and rationalises their actions.
- Relationship between reasoning and rationality on this account: Very tenuous.
 - Reasoning is about producing, assessing and criticising reasons.
 - Rationality is about the (synchronic and diachronic) coherence of one's credences.

"One can be rational without ever reasoning [...]. Conversely, reasoning—even good reasoning—does not necessarily make one rational."
- MacFarlane still talks about both "Bayesian epistemology" and "practical reasoning", but I want to radicalize his ideas in a way that would lead us to reject both terms (at least in the individual case) as chimerical. I want to try out the following radical thesis:

- Credence ascription is only for explaining an individual's actions via decision theory. What rationality requires of one's credences is derived from what rationality requires from the actions they would make one choose. Paradigmatically, a set of credences is irrational if it commits one to falling for a dutch book / dutch strategy. There are no base level epistemic requirements on one's credences.
- Belief, on the other hand, should give credence free reign in an individual's decision-making. In particular, where both doing A and refraining from A is permitted for someone, their act of doing A is never accurately described or explained as the result of practical reasoning. What is normatively required of one's beliefs is derived from what is required from one's behaviour in the social practice of (theoretical) reasoning.

Application to Sleeping Beauty

- The Sleeping Beauty scenario
 - SB is woken up on Monday and Tuesday (with memory loss potion administered in between) if coin lands TAILS
 - SB is woken up only on Monday if coin lands HEADS
 - Upon being woken up, SB is asked: What is the probability of HEADS?
- Applying the above framework:
 - SB's credence in HEADS has to be $1/3$ (otherwise SB is irrational)
 - Why? Dutch Book due to Hitchcock (2004):

Bets	Payoff	Cost	Heads	Tails
bet #1	30 if tails	15	-15	15
bet #2	20 if heads	10	10	-10
bet #3	20 if heads	10 if tails		
only placed if tails		0	-10	
Combined	20 if heads	25 if heads	-5	-5
	30 if tails	35 if tails		

- SB's belief has to be *that* the probability of *HEADS* is $1/2$: She believes this on Sunday and has no reason for switching to any other belief.
- (Darren) Bradley and Leitgeb (2006) take this Dutch Book to show that betting odds and credences come apart in *Sleeping Beauty*.
 - This is another possible conclusion we can draw from it.
 - What I take it to, in any case, show is: In *Sleeping Beauty*, we have to sever either the link between credence and truth, or between credence and decision theory.
 - It seems clear that to a MacFarlanian, only the first option is available. But this means giving up the epistemological significance of credence.
- Different answers to SB force us to give up different attractive principles: Lewis' halfer position violates the Principal Principle. Elga's thirder position respects PP but violates the Reflection Principle and Conditionalisation. My proposed solution avoids such half measures and simply violates all three.